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Bahar

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(54) **ANIONIC ELECTROCHEMICAL
COMPRESSOR AND REFRIGERATION
SYSTEM EMPLOYING SAME**

(58) **Field of Classification Search**

None

See application file for complete search history.

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(US)

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(*) Notice: Subject to any disclaimer, the term of this
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This patent is subject to a terminal dis-
claimer.

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Primary Examiner — Harry D Wilkins, III

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(74) *Attorney, Agent, or Firm* — Invention To Patent
Services; Alex Hobson

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(63) Continuation of application No. 17/751,500, filed on
May 23, 2022, now Pat. No. 11,940,195, which is a
(Continued)

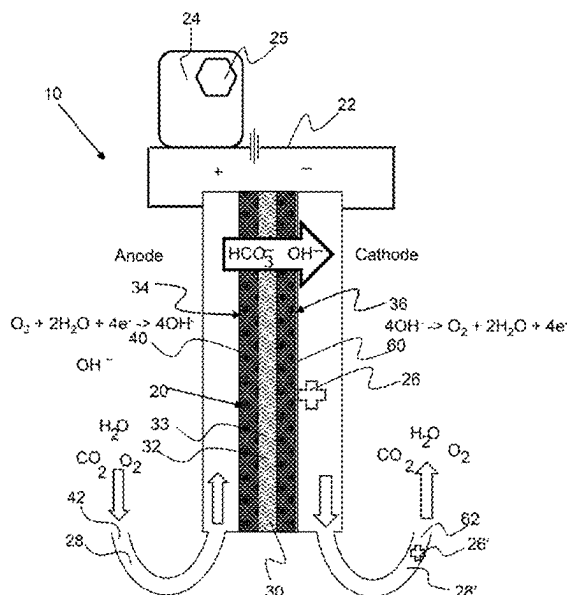
(51) **Int. Cl.**
F25B 9/00 (2006.01)
B01D 67/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **F25B 9/008** (2013.01); **B01D 67/0006**
(2013.01); **B01D 67/00931** (2022.08);
(Continued)

(57) **ABSTRACT**

An electrochemical compressor utilizes an anion conducting layer disposed between an anode and a cathode for transporting a working fluid. The working fluid may include carbon dioxide that is dissolved in water and is partially converted to carbonic acid that is equilibrium with bicarbonate anion. An electrical potential across the anode and cathode creates a pH gradient that drives the bicarbonate anion across the anion conducting layer to the cathode, wherein it is reformed into carbon dioxide. Therefore, carbon dioxide is pumped across the anion conducting layer. The compressor may be part of a refrigeration system that pumps the working fluid in a closed loop through a condenser and an evaporator.

17 Claims, 12 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/800,259, filed on Nov. 1, 2017, now Pat. No. 11,339,996, which is a continuation-in-part of application No. 15/448,734, filed on Mar. 3, 2017, now Pat. No. 11,173,456.

- (60) Provisional application No. 62/303,294, filed on Mar. 3, 2016, provisional application No. 62/416,141, filed on Nov. 1, 2016, provisional application No. 62/430,833, filed on Dec. 6, 2016.

(51) **Int. Cl.**

B01D 69/02 (2006.01)
B01D 69/10 (2006.01)
B01D 69/12 (2006.01)
B01D 71/28 (2006.01)
B05D 3/02 (2006.01)
B05D 3/06 (2006.01)
C08G 61/02 (2006.01)
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C08J 5/22 (2006.01)
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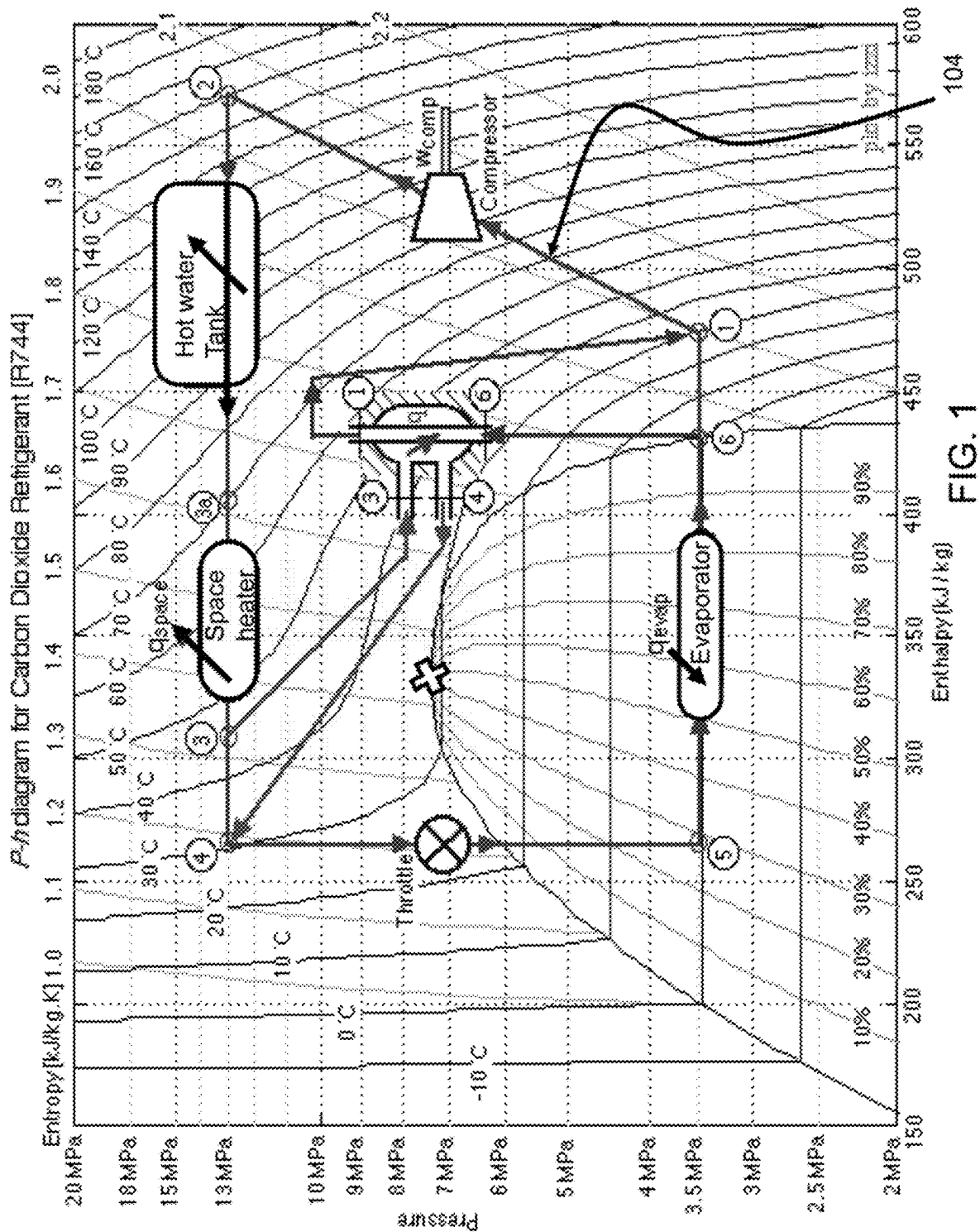
B05D 3/067 (2013.01); **C08G 61/02** (2013.01); **C08J 3/24** (2013.01); **C08J 5/2231** (2013.01); **C08J 5/2256** (2013.01); **C25B 9/23** (2021.01); **F25B 9/002** (2013.01); **F25B 31/00** (2013.01); **B01D 2323/08** (2013.01); **B01D 2323/30** (2013.01); **B01D 2323/345** (2013.01); **B01D 2323/36** (2013.01); **B01D 2325/16** (2013.01); **B01D 2325/42** (2013.01); **C08G 2261/143** (2013.01); **C08G 2261/146** (2013.01); **C08G 2261/312** (2013.01); **C08G 2261/3326** (2013.01); **C08G 2261/516** (2013.01); **C08G 2261/72** (2013.01); **C08J 2325/16** (2013.01); **C08J 2365/02** (2013.01); **Y02E 60/36** (2013.01)

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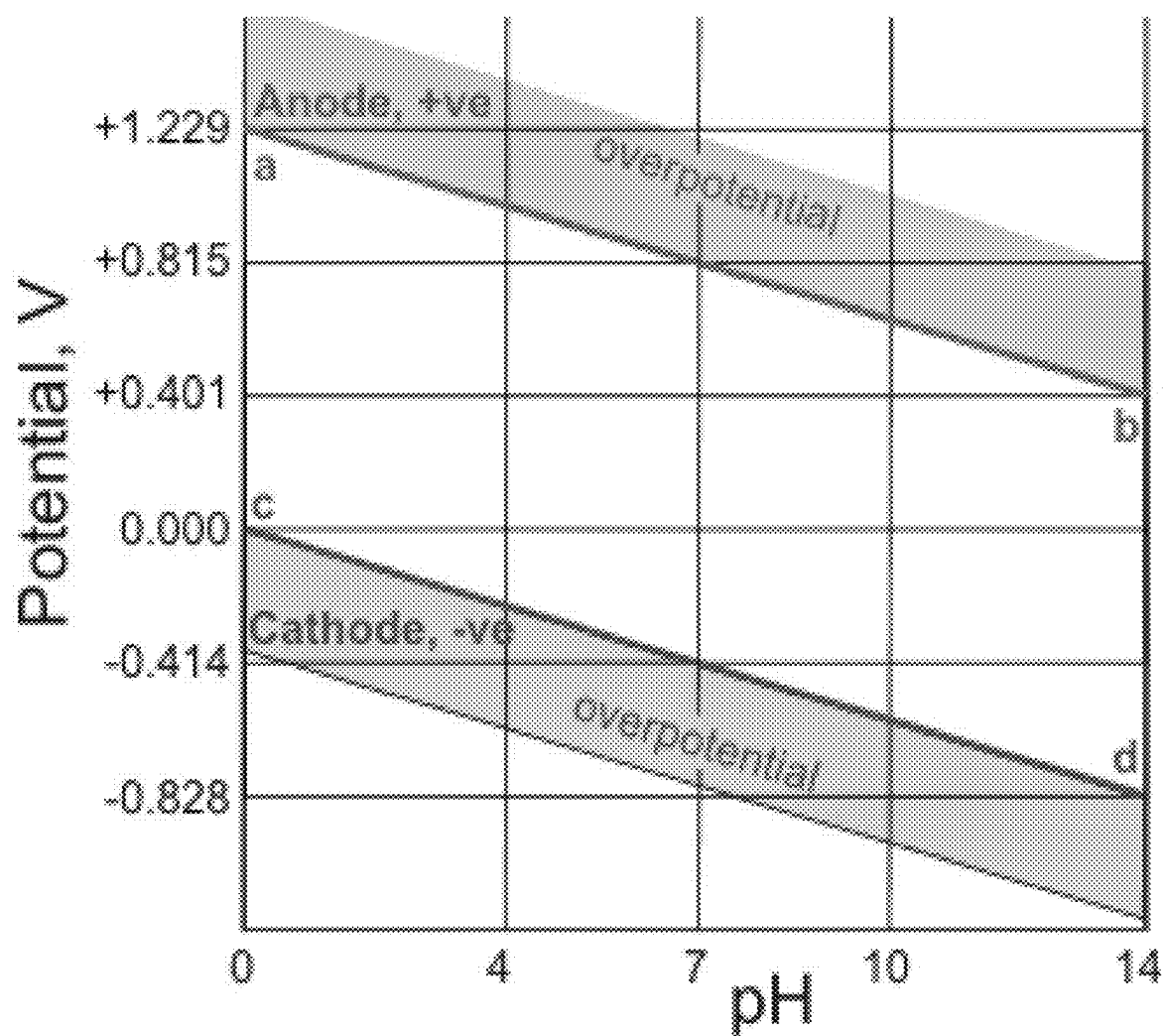


FIG. 2

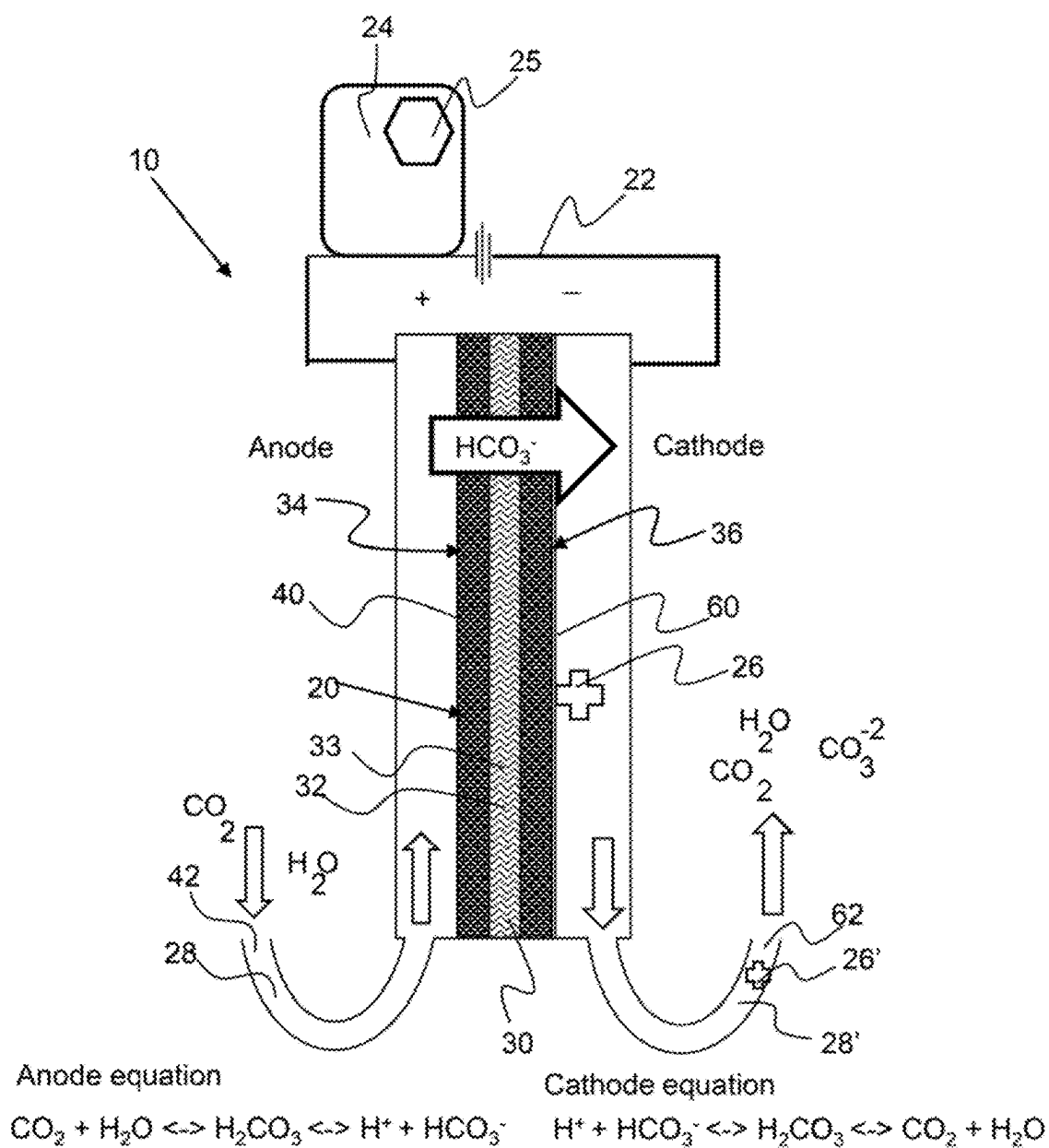


FIG. 3

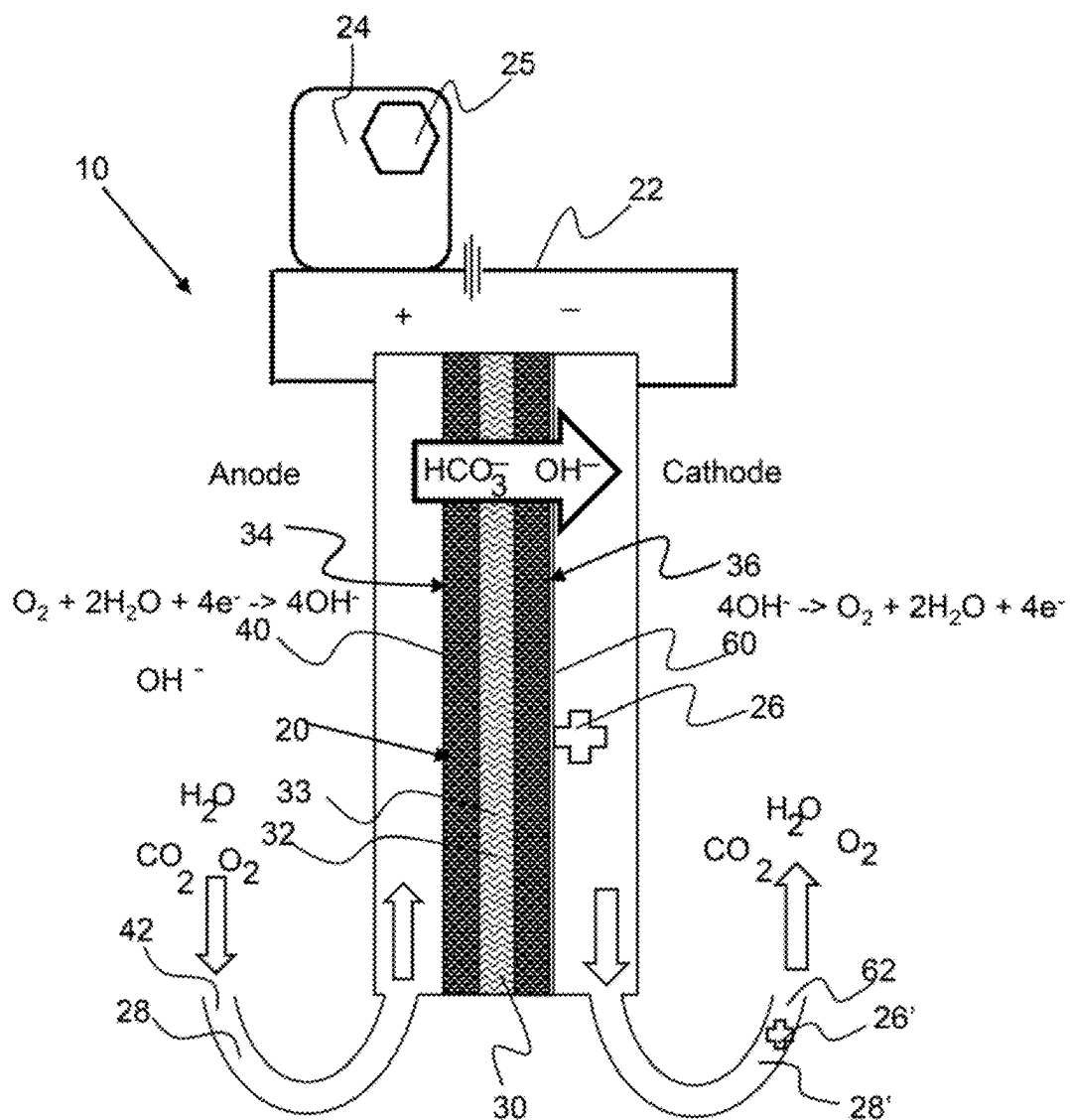


FIG. 4

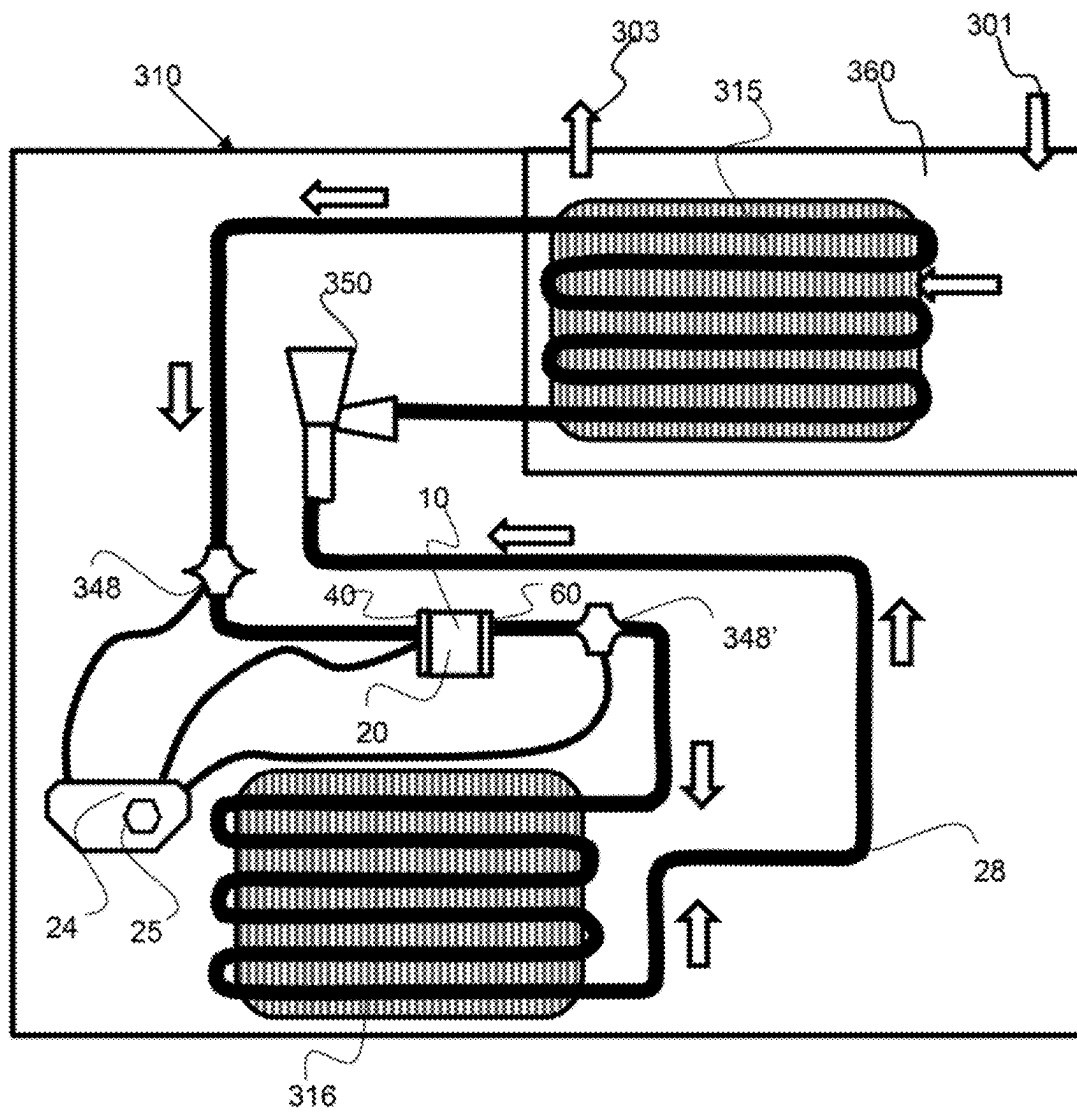


FIG. 5

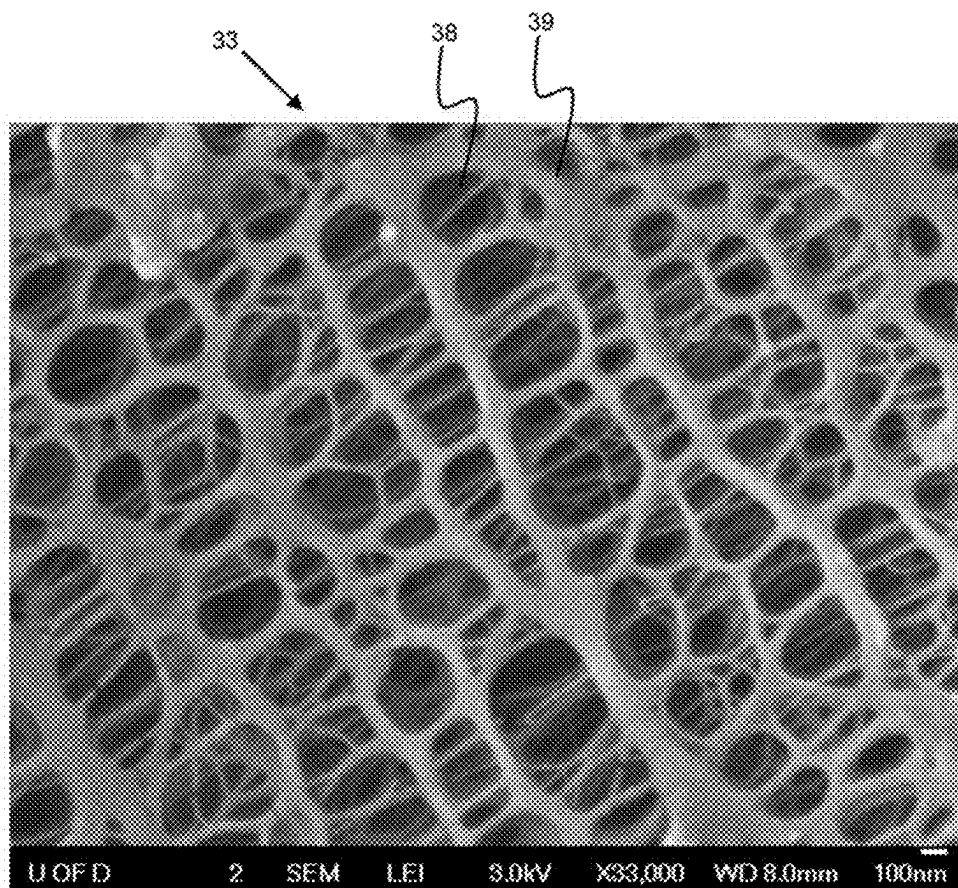


FIG. 6

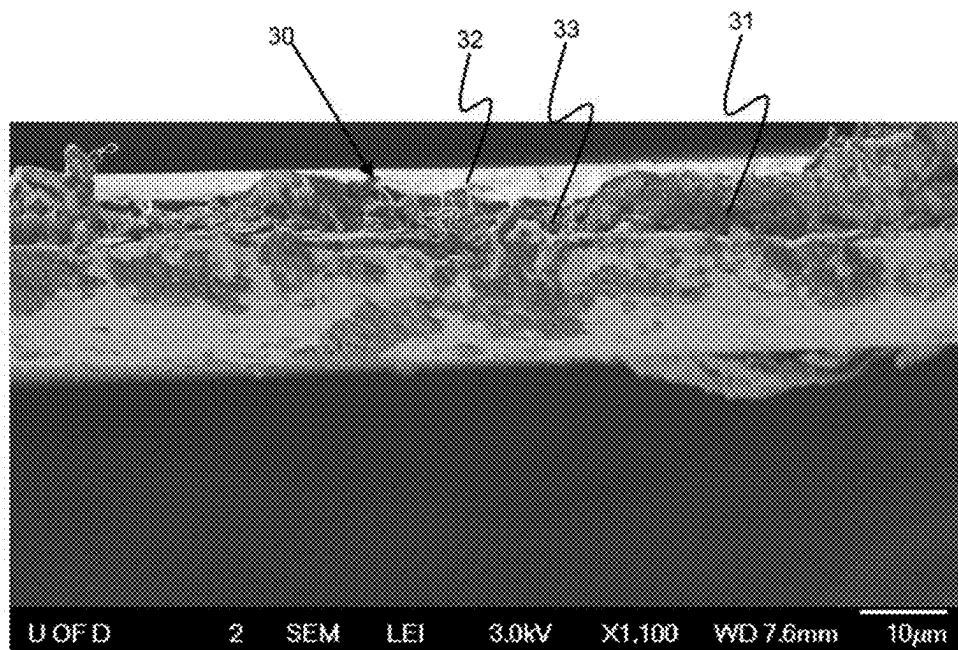
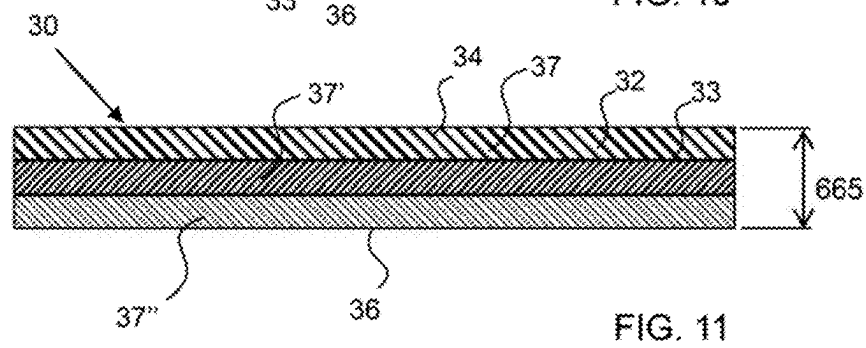
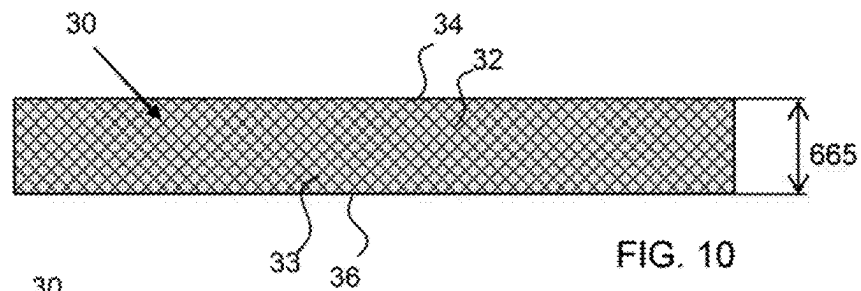
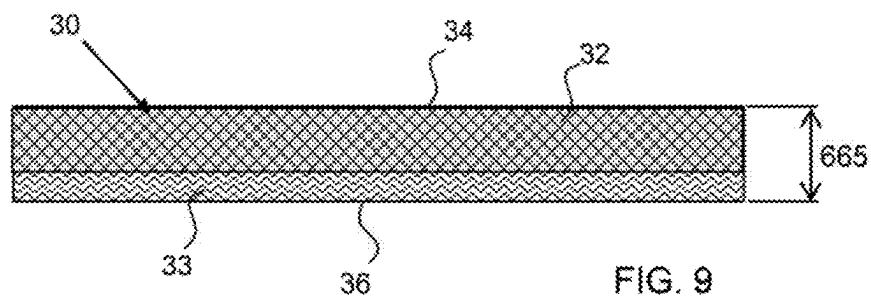
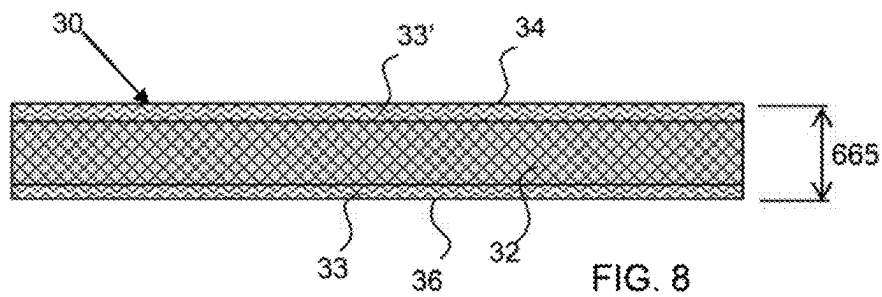


FIG.7



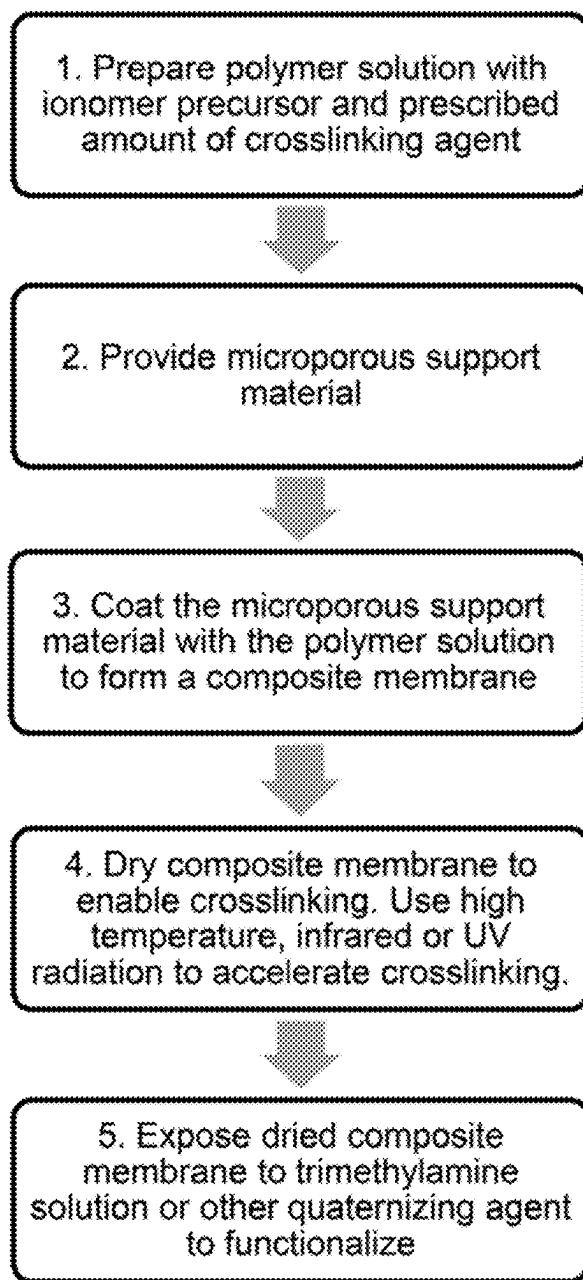


FIG. 12

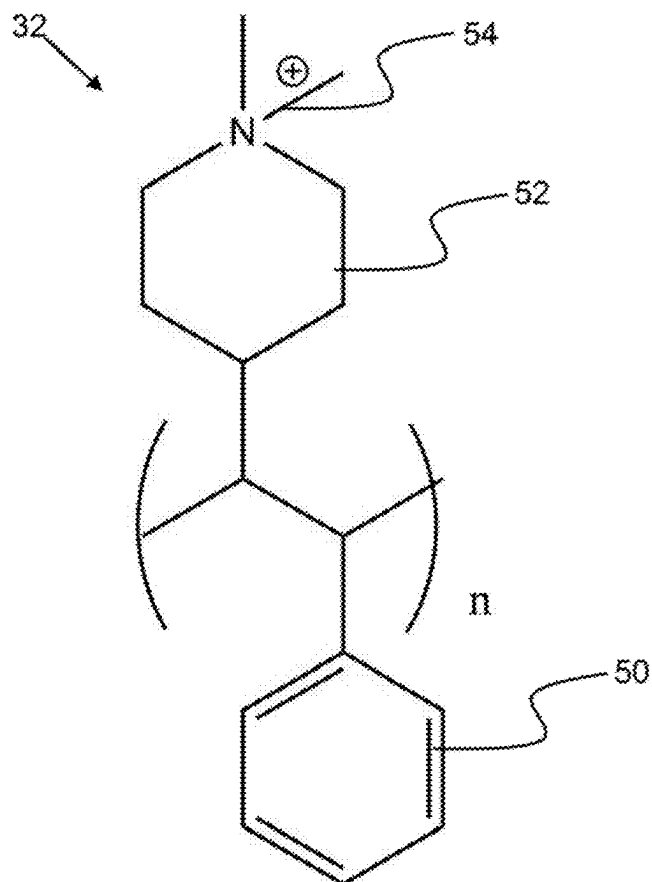


FIG. 13

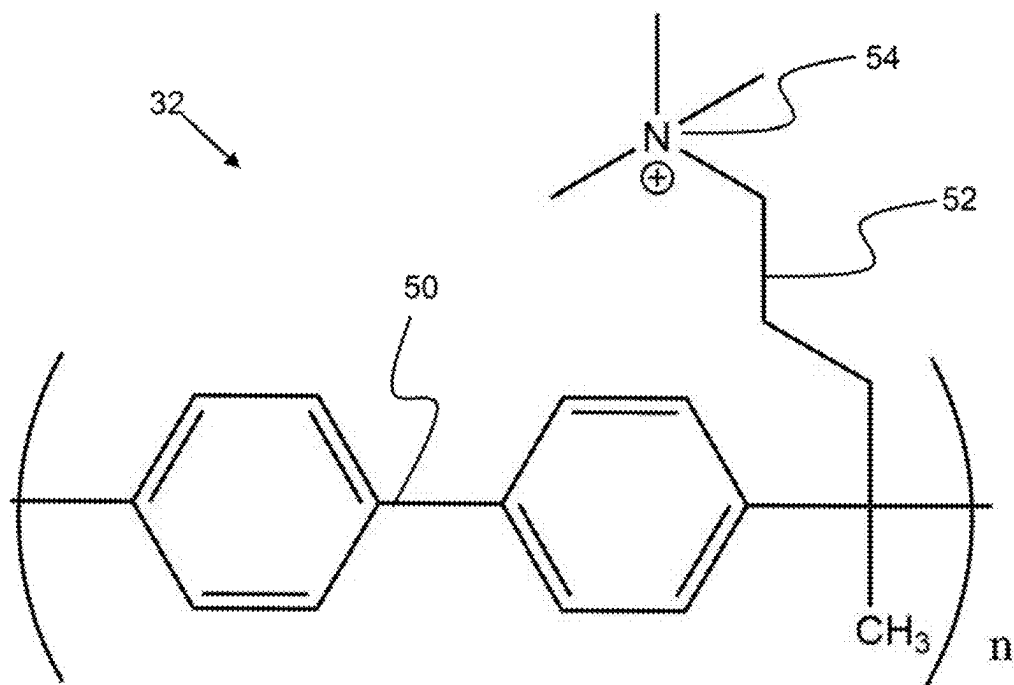


FIG. 14

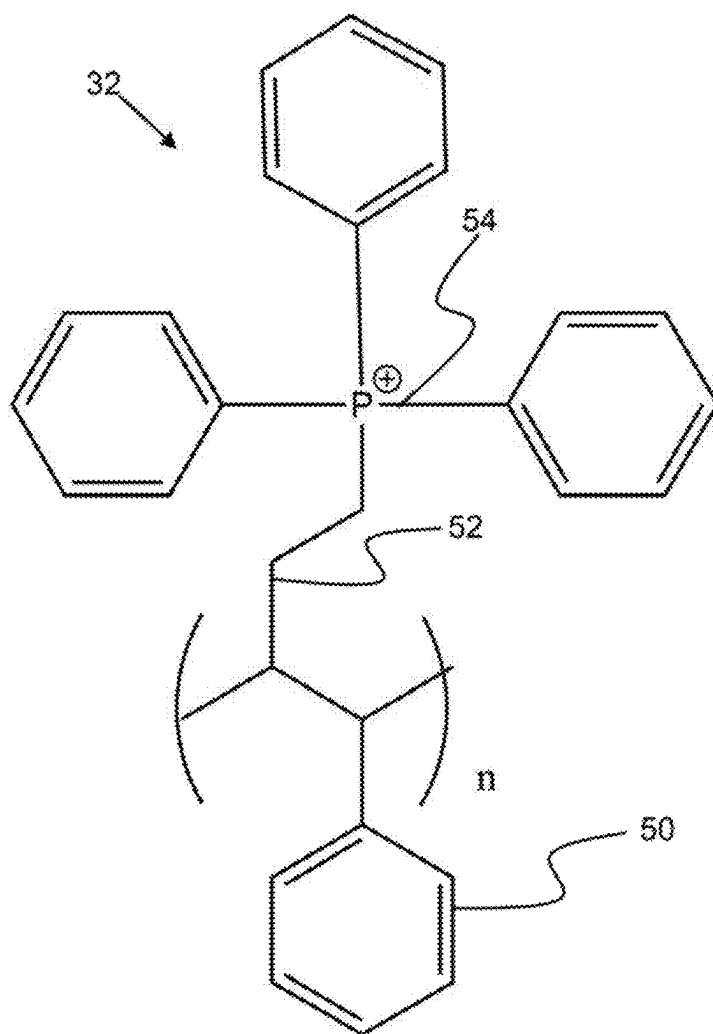


FIG. 15

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ANIONIC ELECTROCHEMICAL COMPRESSOR AND REFRIGERATION SYSTEM EMPLOYING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

The application is a continuation of U.S. patent application Ser. No. 17/751,500 filed on May 23, 2022, which is a continuation of U.S. patent application Ser. No. 15/800,259, filed on Nov. 1, 2017 and issued as U.S. Pat. No. 11,339,996 on May 24, 2022, which is a continuation in part of U.S. patent application Ser. No. 15/448,734, filed on Mar. 3, 2017 and issued as U.S. Pat. No. 11,173,456 on Nov. 16, 2021, which claims the benefit of priority to U.S. provisional application No. 62/303,294, filed on Mar. 3, 2016; and application Ser. No. 15/800,259 claims the benefit of priority to U.S. provisional patent application No. 62/416,141, filed on Nov. 1, 2016, and to U.S. provisional patent application No. 62/430,833, filed on Dec. 6, 2016; the entirety of all former applications are hereby incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

The invention relates to anion electrochemical compressors and refrigeration systems employing said compressor and particularly those that utilize carbon dioxide as a compressible refrigerant.

Background

There is significant global interest in carbon dioxide, CO₂ (R-744), as a refrigerant because of global warming issues concerning current refrigerants and increasing regulations in the United States. Carbon dioxide can offer comparable system efficiency when applied in a subcritical cascade refrigeration cycle, which can be applied in supermarket applications. Interestingly, Coca-Cola has recently announced the conversion of most its beverage dispensing units to carbon dioxide, CO₂ (R-744), based mechanical compression systems. In addition, many U.S. original equipment manufacturers, OEM's, who serve the supermarket refrigeration industry have begun to test R-744 systems. Demonstrating the feasibility of carbon dioxide compression with electrochemical compression technology would have an immediate market opportunity. Unfortunately, the use of carbon dioxide as a refrigerant in mechanical compressors requires high pressures and high compression ratios, such as 100:1, which are difficult to achieve mechanically. Mechanical compressors are noisy and prone to mechanical malfunction and wear.

Developers of mechanical compressors are quick to remind their equipment manufacturers that carbon dioxide compression involves supersonic flow rates, and requires special stainless steel materials to withstand degradation due to the flow rates.

Carbon dioxide is also a fantastic solvent for organic compounds. And this means that every component that meets carbon dioxide needs to be carefully specified to minimize degradation from carbon dioxide contact. This means gaskets, seals, valves, adhesives etc. all need to be carefully selected. It is important to recognize that components used in the system that can operate with carbon dioxide contact cannot be too expensive or impractical for

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commercial integration. Thus, in resolving engineering issues, cost must always be at the forefront of technology decisions being made.

The compression of carbon dioxide is both a technological opportunity and problem. On the one hand, there is a rise in use of carbon dioxide compressors. They are now being widely adopted for heating and cooling applications, in sequestration strategies for carbon dioxide emissions in power plants, in chemical processing using super-critical carbon dioxide as a solvent and in low temperature food and chemical processing. On the other hand, compressing carbon dioxide presents two major problems. First, there is a requirement for high pressure and high compressor temperature and ratios. Second, compressing carbon dioxide is not easy. There is a need for significant improvement in carbon dioxide compression technology.

SUMMARY OF THE INVENTION

The invention is directed to an anion electrochemical compressors and refrigeration systems employing said compressor and particularly those that utilize carbon dioxide as a compressible refrigerant.

Electrochemical compression (ECC) offers a novel route to carbon dioxide compression using novel anionic membranes which would be disruptive and transformational for the global refrigeration industry. Electrochemical compressors are noiseless, solid-state, i.e. no moving parts, modular and scalable. In addition, they are inherently more efficient. Improvements in efficiency would translate immediately to significant reduction in energy use and therefore lower operating costs. In many commercial applications, noise is also a critical issue.

This innovation would expand the use of carbon dioxide compressors to wider scale markets and assist the U.S. in meeting its global obligations. The availability of high performance anionic membranes has been a critical impediment to electrochemical compressor development using anionic chemistry routes. Indeed, electrochemical carbon dioxide compression would have been impossible without the use of these recently developed, advanced membranes.

The present invention provides electrochemical methods to compress carbon dioxide utilizing a pH swing absorption-desorption system. Classically, this could be accomplished by having a sorption/desorption chamber above a caustic solution. The sorption/desorption chamber is connected to a carbon dioxide source. The caustic solution absorbs carbon dioxide then the sorption/desorption chamber is isolated and the caustic solution acidified causing carbon dioxide evolution into the sorption/desorption chamber, which is then connected to a carbon dioxide sink. The process is then repeated to effectively pump carbon dioxide from the source to the sink by shifting the pH of the sorbing/desorbing solution. While this process pumps carbon dioxide, it is very inefficient. The present invention provides for the use an anionic membrane with a significant hydroxide/hydronium gradient imposed on it by an outside electric field to act like the cycling sorption/desorption solution. Hydroxyl ions would migrate to the positive electrode and hydronium ions would migrate to the negative electrode. By putting the hydroxyl rich surface of the membrane in contact with carbon dioxide, carbon dioxide will be expelled from the hydronium rich side of the membrane as any bicarbonate ions and or carbonate ions that diffuse toward that side.

Of importance to the present invention is that the diffusion of the carbonate or bicarbonate ions through the membrane and to the acid rich side will be insufficient to generate a

significant flux of evolving carbon dioxide and the mechanical integrity of the anionic membranes to withstand the pressure. A sufficient gradient to absorb carbon dioxide from the source will be required.

Electrochemical compression and hydrogen pumping all are based on active electrochemistry. In hydrogen pumping, the hydrogen is oxidized at the anode to protons which migrate through a cationic membrane, such as perfluoro-sulfonic acid, are reduced on the cathode. In the present invention, the carbon dioxide pumped is not via reduction and oxidation so it is significantly different.

The compression of carbon dioxide is both a technological opportunity and problem. On the one hand, there is a rise in use of carbon dioxide compressors. They are now being widely adopted for heating and cooling applications, in sequestration strategies for carbon dioxide emissions in power plants, in chemical processing using super-critical carbon dioxide as a solvent, in low temperature food and chemical processing. On the other hand, compressing carbon dioxide presents some challenges. There is a requirement for high pressure and high compressor temperature and ratios. There is a need for significant improvement in carbon dioxide compression technology.

Carbon dioxide is a high-pressure refrigerant where high operating pressures are required for efficient operation. During non-compression phases of mechanical compressors, the ambient temperature can exceed the critical temperature and the pressure can exceed the critical pressure. Hence systems are typically designed to withstand pressures up to 90 bar, or sometimes even equipped with a small standstill condensing units to keep pressures low. At the same time, carbon dioxide has a low compression pressure ratio (20 to 50% less than HFCs and ammonia), which improves volumetric efficiency. With evaporation temperatures in the range of -55°C . to 0°C ., the volumetric performance of carbon dioxide is, for example, four to twelve times better than that of ammonia which allows compressors with smaller swept volumes to be used.

The triple point and critical point of carbon dioxide are very close to the working range. The critical point may be reached during normal system operation. During system service, the triple point may be reached, as indicated by the formation of dry ice when liquid containing parts of the systems are exposed to atmospheric pressure. Special procedures are necessary to prevent the formation of dry ice during service venting.

The reason that existing carbon dioxide compressor designs are so expensive is, in part, because the overall pressure ratio is 100:1, and, in part, because carbon dioxide requires stainless steel construction in the presence of water vapor. But by far, the most significant impact on cost is an aerodynamic design practice that limits the design pressure ratio per stage on heavier gases such as carbon dioxide.

Standard turbomachinery design practice is to limit the inlet flow Mach number to less than 0.90 at the inducer blade tip to avoid generating shock waves in the blade passages and their accompanying losses. This is typically done by adjusting the stage speed. The Mach number itself is a function of molecular weight and therefore the effect is more pronounced on the heavier-than-air carbon dioxide. This inducer blade tip speed limit results in a pressure ratio per stage limits of approximately 1.7 to 2.0:1 on carbon dioxide. At these stage pressure ratios, eight stages of compression are typically required to reach an overall pressure ratio of 100:1

Electrochemical Compression

An anionic membrane is used with a significant hydroxide/hydronium ion gradient imposed on it by an outside electric field to act like the cycling sorption/desorption solution. Hydroxyl ions would migrate to the positive electrode and hydronium ions would migrate to the negative electrode. By putting the hydroxyl rich surface of the membrane in contact with carbon dioxide the carbon dioxide is expelled from the hydronium rich side of the membrane as any bicarbonate and or carbonate ions diffuse toward that side.

Ion-Exchange Membranes

At the heart of an electrochemical compression is an ion-exchange membrane with two catalytic electrodes. The entire assembly is referred to as a membrane-electrode assembly (MEA). Ion-exchange membranes transport ions across an ion conductive polymeric membrane, often referred to as an 'ionomer'. Ion-exchange membranes are made of a polymeric material attached to charged ion groups. Anion-exchange membranes contain fixed cationic groups, such as ammonium or phosphonium, with mobile anions, which provide the ionic conductivity. Cation-exchange membranes contain fixed anionic groups, such as carboxylic or sulfonic acid, with mobile cations, which provide the conductivity. The selectivity of the membranes is due to Donnan equilibrium and Donnan exclusion and not due to physically blocking or electrostatically excluding specific charged species.

Anion Exchange Membranes

Of particular interest for electrochemical carbon dioxide compressors are anion exchange membranes or anion conducting layer. Electrochemical cells employing anion exchange membranes may function without the use of platinum-group metal catalysts or acid-resistant metal housings, significantly lowering their cost. Additionally, the ability to transport anions, such as hydroxide or bicarbonate, has great utility in electrochemical carbon dioxide compression.

An exemplary anion conducting polymer comprises quaternary ammonium or phosphonium functional groups, with poly(styrene), poly(phenylene), or poly(arylene) backbones. Rigid, aromatic polymer backbones such as poly(phenylene) or poly(arylene) provide high tensile strength along with resistance to chemical degradation via hydroxide elimination reactions in a highly caustic environment. Ion exchange membranes produced with these ionomers can further be reinforced by porous support materials, such as microporous polytetrafluoroethylene, polyethylene, or polypropylene membranes. Reinforcing the ionomer with the porous support matrix creates a composite anion exchange membrane. The preferred microporous support for use in the present invention is porous ultra-high molecular weight polyethylene, as it has superior chemical compatibility (compared to expanded polytetrafluoroethylene, the standard for reinforced cation exchange membranes) and porosity (compared to polypropylene, an alternative polyolefin support). An exemplary ion exchange membrane for use in the present invention comprises a polymer with a poly(arylene) or poly(phenylene) backbone and alkyl or piperidine side chains featuring quaternary ammonium or phosphonium groups for ionic conductivity. In an exemplary embodiment, a solution of this ionomer is impregnated into a microporous polyolefin support for greater reinforcement and stability, especially at lower thickness.

An exemplary anion conducting layer is a composite anion conducting layer comprising an anion conducting polymer that is reinforced by a support material. An even more desirable example of the present invention involves

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impregnating a microporous polyolefin support material between 5 and 50 microns, with porosity ranging from approximately 50% to 90% and pore size between approximately 20 nm and 1 micron, with a polymer solution comprising a precursor form of the ionomer comprising tertiary amine groups grafted to a poly(arylene) or poly(phenylene) backbone, along with a crosslinking agent such as divalent metal cations, tetramethyl-1,6-hexanediamine, or 4,4'-trimethylenebis(1-methyl-piperidine), and then exposing the dried composite membrane to trimethylamine solution in water or ethanol. The crosslinking can be initiated or accelerated by exposure to high temperatures as well as infrared or ultraviolet radiation.

An exemplary anion conducting layer is an anisotropic anion conducting layer, that has varying properties through the thickness of the layer, and may comprise a series of thin films fused together to create an anisotropic membrane. Typically, quaternary ammonium ions are the cationic site and the backbone is varied, however it is possible to create cationic species with phosphonium as the cationic center. The number of layers can be altered as well as step changes in the blend ratio to generate membranes of significantly anisotropic internal structures.

The anion conducting polymer within an anion conducting layer may be crosslinked using a crosslinking agent or compound. Anion conducting polymers, such as within a composite anion conducting layer may be crosslinked to increase their mechanical and chemical stability, especially in hydrated conditions. In the case of an anionic ionomer with functional quaternary ammonium groups, crosslinks may be made between polymer chains by linking quaternary ammonium groups together with crosslinking agents such as polyamines, blocked polyamines, dicyanodiamides, divalent metal cations, tetramethyl-1,6-hexanediamine, 4,4'-trimethylenebis(1-methyl-piperidine), or 4,4'-(1,3-Propanediyl)bis(1-methyl-piperidine). A composite anion conducting layer may be formed by imbibing a support material with a polymer solution containing the ionomer along with one of the above crosslinking agents at a prescribed molar ratio of crosslinking agent to functional ionic groups. Any suitable anion conducting membrane may be used in the present invention. These membranes are characterized by nano-scale channels that essentially hold water and conduct anions (such as hydroxyl ions). These new anion exchange membranes have demonstrated the ability to achieve high conductivity for anions or high permselectivity.

Electrochemical Compression of Carbon Dioxide

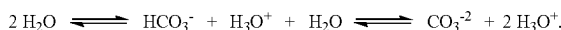
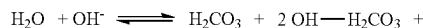
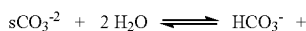
Carbon dioxide reacts with water. The carbon atom of carbon dioxide is electron poor with an oxidation state of IV. The electron rich oxygen of water donates an electron pair to the carbon. After proton transfer from water to an oxygen of the carbon dioxide unit, carbonic acid is formed as shown in by equation:



The reaction between water and dissolved carbon dioxide is reversible and rapid. Carbonic acid is in equilibrium with the bicarbonate anion. The bicarbonate ion (hydrogen-carbonate ion) is an anion with the empirical formula HCO_3^- and a molecular mass of 61.01 Daltons; it consists of one central carbon atom surrounded by three oxygen atoms in a trigonal planar arrangement, with a hydrogen atom attached to one of the oxygens. It is isoelectronic with nitric acid HNO_3 . The bicarbonate ion carries a negative one formal

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charge and is the conjugate base of carbonic acid H_2CO_3 ; at the same time, it is the conjugate acid of CO_3^{2-} , the carbonate ion, as shown by these equilibrium reactions:

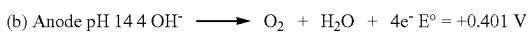
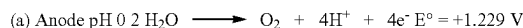


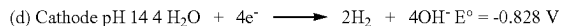
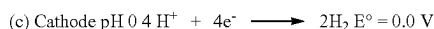
The hydration equilibrium constant at 25° C. is called K_h , which in the case of carbonic acid is $[\text{H}_2\text{CO}_3]/[\text{CO}_2] \approx 1.7 \times 10^{-3}$ in pure water. Hence, most the carbon dioxide is not converted into carbonic acid, remaining as CO_2 molecules. In the absence of a catalyst, the equilibrium is reached quite slowly. The rate constants are 0.039 s^{-1} for the forward reaction ($\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$) and 23 s^{-1} for the reverse reaction ($\text{H}_2\text{CO}_3 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$). The addition of two molecules of water to CO_2 would give orthocarbonic acid, $\text{C}(\text{OH})_4$, which exists only in minute amounts in aqueous solution.

Addition of base to an excess of carbonic acid gives bicarbonate anions, (hydrogen carbonate). With excess base, carbonic acid reacts to give carbonate salts. Bicarbonate ions (HCO_3^-) are anions and therefore can travel through an appropriate electrolyte, such as an aqueous solution or solid polymer anion exchange membrane.

When an electric potential is established through water, it causes positive ions, including the inherent hydrogen ions H_3O^+ , to move towards the negative electrode, cathode, and negative ions, including the inherent hydroxide ions OH^- , to move towards the positive electrode, anode. This is the nomenclature conventionally used in an electrolytic cell. At sufficient potential difference, this may cause electrolysis with oxygen gas being produced at the anode and hydrogen gas produced at the cathode. Since we do not want hydrogen or oxygen to evolve, this system will aim to operate with minimal potential difference—which will also provide lowest energy utilization i.e. high operating efficiency.

Generally, the water next to the electrodes will change pH due to the ions produced or consumed. If the electrode compartments are separated by a suitable porous membrane then the concentration of H_3O^+ in the anolyte and OH^- in the catholyte, and hence the increase in the respective conductivities are both expected to increase more than if there is free mixing between the electrodes, when most of these ions will neutralize each other. Small but expected differences in the solutions' pH next to the anode, anolyte, and cathode, catholyte, cause only a slight change to the overall potential difference required (1.229 V). Increasing the acid content next to the anode due to the H_3O^+ produced will increase its electrode potential (for example: pH 4 $E = +0.992 \text{ V}$) and increasing the alkaline content next to the cathode due to the OH^- produced will make its electrode potential more negative (for example: pH 10 $E = -0.592 \text{ V}$). If the anode reaction is forced to run at pH 14 and the cathode reaction is run at pH 0 then the electrode potentials are +0.401 V and 0 V respectively (see above right).





Thus, by utilizing an electrochemical system, the dissolution of carbon dioxide into water is enhanced and equilibrium is shifted towards increased production of carbonic acid and bicarbonate with the presence of OH ions at the anode. The carbonic acid or bicarbonate ions are then split into carbon dioxide and water at the cathode with the presence of H⁺ ions (protons), and high pressure carbon dioxide is released. Temperature effects should also be considered. Since the carbon dioxide entering the anode compartment will be coming from an evaporator, it will have a low temperature (i.e. solubility will be enhanced), and excitation with protons at the cathode will enhance carbon dioxide evolution. The electrochemical process therefore accelerates the natural process of carbon dioxide dissolution and evolution. It aids the process.

Carbon Dioxide—Carbonic Acid Equilibrium

Carbonic acid is formed when atmospheric carbon dioxide is dissolved in water. The chemical equilibria are:

- 1) Gas dissolution;
- 2) Carbonic acid formation; and
- 3) Carbonic acid equilibrium

In an open system, the partial pressure of CO₂ (g) is relatively constant at P (CO₂)=0.000355 Atmosphere. The solubility of gases decreases with increasing temperature. The electron rich oxygen of water donates an electron pair to the carbon. After proton transfer from water to an oxygen of the CO₂ unit, carbonic acid is formed. The reaction between water and dissolved carbon dioxide is reversible and rapid. Carbonic acid is in equilibrium with the bicarbonate anion.

Carbonic acid is a chemical compound with the chemical formula H₂CO₃ (equivalently CO(OH)₂). It is also a name sometimes given to solutions of carbon dioxide in water (carbonated water), because such solutions contain small amounts of H₂CO₃. In physiology, carbonic acid is described as volatile acid or respiratory acid, because it is the only acid excreted as a gas by the lungs. It plays an important role in the bicarbonate buffer system to maintain acid-base homeostasis.

Electrodes

Standard electrodes for Water electrolysis can be used, since we are essentially simply driving a potential across the cell to enhance carbonic acid formation and dissociation. However, there is literature that suggest that carbonic acid formation can be further enhanced with the presence of high surface area catalysts containing copper and/or palladium.

It is anticipated that once the cell is operating without catalyst that further enhancements to operating rates will be investigated with the use of catalysts. Xergy has produced electrodes for water electrolysis for its own water electrolyzer program.

Ammonia as a Working Fluid

A similar approach can be taken by using Ammonia as a working fluid. Similarly, a chemical reaction also occurs when ammonia dissolves in water. In aqueous solution, ammonia acts as a base, acquiring hydrogen ions from H₂O to yield ammonium and hydroxide ions. In contrast, the ammonium ion acts as a weak acid in aqueous solution

because it dissociates to form hydrogen ion and ammonia. Therefore, in a similar manner to CO₂, adsorption and desorption with the aid of an electrochemical cell pumps ammonia from one side to the other, and thus enables ammonia compression.

This application incorporates by reference in their entirety, the following applications: U.S. application Ser. No. 15/289,220 filed on Oct. 10, 2016, entitled Electrochemical Heat Transfer System and currently pending, which is a continuation of U.S. application Ser. No. 13/029,006 filed on Feb. 16, 2011, entitled “Electrochemical Heat Transfer System” and issued as U.S. Pat. No. 9,464,822 on Oct. 11, 2016, which is a continuation-in-part of U.S. application Ser. No. 12/771,620 filed on Apr. 30, 2010, entitled “Self-Contained Electrochemical Heat Transfer System” and issued as U.S. Pat. No. 8,627,671 on Jan. 14, 2014, which claims the benefit of U.S. provisional patent application No. 61/215,131 filed on May 1, 2009, and U.S. application Ser. No. 13/029,006 claims priority to U.S. provisional application No. 61/305,410 filed on Feb. 17, 2010 and entitled “Electrochemical Heat Pump System for Cooling Electronic Components,” and to U.S. provisional application No. 61/347,428, filed May 23, 2010 and entitled “Compact Cooling Systems Using Electrochemical Compression,”; U.S. provisional application No. 62/262,845, filed on Dec. 3, 2015, and entitled “Electrochemical Compressor and Refrigeration System Using Same”; U.S. patent application Ser. No. 14/859,267 filed on Sep. 19, 2015 and entitled “Electrochemical Compressor Based Heating Element and Hybrid Hot Water Heater Employing Same”, which is a continuation in part of U.S. patent application Ser. No. 13/899,909 filed on May 22, 2013, entitled “Electrochemical Compressor Based Heating Element And Hybrid Hot Water Heater Employing Same” which is now abandoned; U.S. provisional application No. 62/262,845, filed on Dec. 3, 2015, entitled “Electrochemical Compressor and Refrigeration System Using Same”; U.S. patent application Ser. No. 13/725,515 filed on Dec. 21, 2012 and currently pending; Ser. No. 14/859,267 filed on Sep. 19, 2015 and currently pending, Ser. No. 13/943,619 filed on Jul. 16, 2013 and currently pending, Ser. No. 14/630,659 filed on Feb. 25, 2015 and currently pending, Ser. No. 14/712,376 filed on May 14, 2015 and currently pending, and Ser. No. 15/173,854 filed on Jun. 6, 2016 and currently pending; the entirety of all related applications are hereby incorporated by reference herein.

The summary of the invention is provided as a general introduction to some of the embodiments of the invention, and is not intended to be limiting. Additional example embodiments including variations and alternative configurations of the invention are provided herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate embodiments of the invention, and together with the description serve to explain the principles of the invention.

FIG. 1 shows a pressure enthalpy chart and a critical point and the extent of the transcritical fluid region.

FIG. 2 is a graph of potential versus pH.

FIG. 3 shows a schematic of a carbon dioxide electrochemical compressor.

FIG. 4 shows a schematic of a carbon dioxide electrochemical compressor.

FIG. 5 shows a schematic of an exemplary refrigerant system employing an electrochemical carbon dioxide compressor.

FIG. 6 is a scanning electron microscopy (SEM) image of the face of a microporous polypropylene membrane support.

FIG. 7 is a scanning electron microscopy (SEM) image of the cross-section of a composite anion exchange membranes suitable for use in the invention.

FIGS. 8 to 11 show a cross-sectional schematic of anisotropic composite ion exchange membranes.

FIG. 12 shows a flow diagram explaining the method of preparing a composite crosslinked anion exchange membrane.

FIGS. 13 to 15 show molecular diagrams for an exemplary anion conducting polymers having a backbone, a side chain and a functional group.

Corresponding reference characters indicate corresponding parts throughout the several views of the figures. The figures represent an illustration of some of the embodiments of the present invention and are not to be construed as limiting the scope of the invention in any manner. Further, the figures are not necessarily to scale, some features may be exaggerated to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Also, use of “a” or “an” are employed to describe elements and components described herein. This is done merely for convenience and to give a general sense of the scope of the invention. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

Certain exemplary embodiments of the present invention are described herein and are illustrated in the accompanying figures. The embodiments described are only for purposes of illustrating the present invention and should not be interpreted as limiting the scope of the invention. Other embodiments of the invention, and certain modifications, combinations and improvements of the described embodiments, will occur to those skilled in the art and all such alternate embodiments, combinations, modifications, improvements are within the scope of the present invention.

FIG. 1 shows a pressure enthalpy diagram for carbon dioxide refrigerant and a critical point and the extent of the transcritical fluid region. Notice that the compressor does not follow an isentropic process as shown by line 104. This is a practical system with data adapted from a Visteon Corp. automobile a/c design.

FIG. 2 is a graph of potential versus pH and shows how the potential of the anode and cathode can impact the pH of water.

Referring now to FIGS. 3 and 4, an exemplary carbon dioxide electrochemical compressor 10 pumps carbon dioxide from the anode 40 to the cathode 60. The exemplary carbon dioxide electrochemical compressor comprises a

power supply 22 connected to a membrane electrode assembly 20 comprising an anode 40 and cathode 60 separated by an anion conducting layer 30. The anion conducting layer 30 may comprise an anion conducting polymer 32 and a support material 33, such as a porous polymer material. dioxide from the anode to the cathode. As shown in FIG. 3, the carbon dioxide gas is dissolved in the water, and is partially converted to carbonic acid, which is in equilibrium with bicarbonate anion as shown by the anode equation. The power supply produces a voltage potential between the anode and cathode to change the pH as shown in FIG. 2. The bicarbonate is driven across the anion conducting layer 30 by the gradient in pH from the anode to the cathode and is mobile inside the anion exchange membrane, and diffuses or is transported across the membrane, as shown in FIG. 3. The bicarbonate reacts on the cathode to reform carbon dioxide as shown by the Cathode equation in FIG. 3. As shown in FIG. 4, hydroxyl ions are depleted which drives the equilibrium to the right, or to the formation of carbon dioxide. As shown in FIG. 4, hydroxyl ions are produced at the anode driving the equilibrium to the formation of bicarbonate. The hydroxyl ion formation and depletion are controlled by the potential produced by the power supply across the anode and cathode in the presence of oxygen. The working fluid comprises water, carbon dioxide and oxygen. Therefore, in essence, carbon dioxide is pumped from the anode to the cathode. The voltage potential from the anode to the cathode is provided by a power source and may be controlled by a controller 24 having a microprocessor 25. The controller 24 may control the voltage potential based on a desired or set flow rate of the working fluid as measured by a sensor 26'. Any number of sensors 26, 26' may be configured to measure the flow rate, pH of the working 28, temperature, pressure on the anode or cathode side and the like. A sensor 26 may provide data to the controller and the controller may control the potential across the anode and cathode as a function of the data input from the sensor. The voltage may be oscillated or maintained at a set value for a period of time. The anode may have an input 42 for working fluid 28 and the cathode may have an outlet 62 for the working fluid 28' at a higher pressure than at the anode.

As shown in FIG. 5, an exemplary carbon dioxide electrochemical compressor 10 is configured in a refrigeration system 310 that pumps the working fluid 28, water and carbon dioxide, in a closed loop from the anode 40, through the membrane electrode assembly 20, to the cathode 40 and back to the anode. The refrigeration system may comprise a condenser 316, an evaporator 315 and an expansion valve 350. A controller 24 having a microprocessor 25 may control the voltage across the anode and cathode to control the rate of flow of working fluid through the refrigerant system. The system may have a plurality of valves 348 to control flow and pressure of the working fluid. The refrigerant system may cool a flow of fluid, such as air, that has an inlet 301 and outlet 303 across the evaporator to cool the fluid from the inlet to the outlet. The evaporator may be in an exchange chamber 360.

FIG. 6 shows a scanning electron microscopy image of a support material 33, which is a microporous polypropylene support having pores 38 that extend through the thickness of the layer. The nanoscale morphology of the support material allows hydrophobic domains of the ionomer to segregate around the fibers 39 of the porous support, creating hydrophilic channels for water and anions to flow through.

FIG. 7 shows a scanning electron microscopy image of the cross-section of an anion conducting layer 30, that is a composite anion exchange membrane 31, having a thickness

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of no more than about 50 microns, and preferably no more than about 25 microns. The composite anion exchange membrane 31 comprises a support material 33, and an anion conducting polymer 32 attached thereto and therein. Note that the ionomer appears to be well-imbibed; very little, if any, of the porous support material morphology is visible from the cross-section.

FIGS. 8 to 11 shows examples of anion conducting layers 30. The layers have a thickness 665 and an anion conducting polymer 32 that may be coupled with a support layer 33. The support layer may extend to both the anode side 34 and cathode side 36. The support layer may extend completely through from the anode to the cathode side, as shown in FIG. 8, or may be only on one side, such as the cathode side 36, as shown in FIG. 9. The anion conducting polymer 32 may extend through the support material 33 to both the anode and cathode side as shown in FIG. 10. The anion support material may have a thickness extending from the support material, or buttercoat thickness on the anode and/or cathode side. As shown in FIG. 11, an anion conducting layer 30 may comprise a gradient of properties through the thickness, wherein the properties change from the anode to the cathode side. A layer of anion conduction polymer 37, or layer exposed to the anode, may have high concentration of functional groups, such as quaternary ammonium groups, wherein the mol percent as a function of the anion conducting polymer is high as compared with the mol percent of the anion conducting polymer in the cathode layer 37', or layer exposed to the cathode. The anode layer 37 may be separated from the cathode layer 37' by middle layer 37''. The middle layer may have a concentration of functional groups that is between the concentration of the functional groups of the anode layer and the cathode layer. The layers may be formed together or may be separate layers that are bonded together or attached to form the anisotropic anion conducting layer 30. An anisotropic anion conducting layer may have a support material 33, in one or more of the layers and these layers may have a different concentration of functional groups or different types of functional groups. The concentration gradient of the functional groups in the anion conducting polymer may vary through the thickness of the anion conduction layer 30 by as much as 10 to 1 or more, or about 5 to 1 or more, or about 2 to 1 or more, or about 1.5 to 1 or more and any range between and including the concentration gradient ratios provided. A first layer of an anisotropic anion conducting layer may comprise a concentration of quaternary ammonium groups and a second layer may comprise a concentration of phosphonium groups. The layers may comprise only one type of functional group or a blend of two or more functional groups.

FIG. 12 is a flowchart explaining the method for preparing a composite anion exchange membrane relating to the method claims. As detailed in the diagram, a composite anion conducting layer may be formed by coating or impregnating a porous support material with an anion conducting polymer. The coated anion conducting polymer may at least partially or fully fill the porosity or pores of the porous support material. The polymer solution that is coated in or onto the porous support material may comprise the anion conducting polymer as well as a cross-linking agent. After coating the porous substrate with the polymer solution, the polymer solution may be crosslinked by an increase in temperature or exposure to infrared or ultraviolet (UV) radiation to form a crosslinked polymer. The crosslinked polymer may then be functionalized by the exposure to quaternizing agent to form a composite anion conducting layer.

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FIGS. 13 to 15 show molecular diagrams for an exemplary anion conducting polymers having a backbone, a side chain and a functional group. FIG. 13 shows an exemplary anion conducting polymer having a quaternary ammonium functional group 54, a piperidine side chain 52, and a poly(styrene) backbone 50. FIG. 14 shows an exemplary anion conducting polymer having a quaternary ammonium functional group 54, an alkyl side chain 52 and a poly(arylene) or poly(phenylene) backbone 50. FIG. 15 shows an exemplary anion conducting polymer having a phosphonium functional group 54, an alkyl side chain 52 and a poly(styrene) backbone 50.

It will be apparent to those skilled in the art that various modifications, combinations and variations can be made in the present invention without departing from the spirit or scope of the invention. Specific embodiments, features and elements described herein may be modified, and/or combined in any suitable manner. Thus, it is intended that the present invention cover the modifications, combinations and variations of this invention provided they come within the scope of the appended claims and their equivalents.

What is claimed is:

1. A membrane electrode assembly comprising:

- a) an anode;
- b) a cathode; and
- c) an anisotropic anion conducting layer between and in contact with the anode and cathode and comprising an anion conducting polymer;

wherein the anion conducting polymer comprises a backbone selected from the group consisting of: poly(arylene), poly(phenylene) and poly(styrene); and wherein the anisotropic anion conducting layer has a concentration gradient of the functional groups from the anode to the cathode of at least 1.5:1.

2. The membrane electrode assembly of claim 1, wherein the anisotropic anion conducting layer comprise a first anion conducting layer having a first concentration of functional groups and a second anion conducting layer having a second concentration of functional groups and wherein the first anion conducting layer is in contact with the anode and wherein the second anion conducting layer is in contact with the cathode.

3. The membrane electrode assembly of claim 2, wherein the first concentration of functional groups is at least 1.5 greater than the second concentration of functional groups.

4. The membrane electrode assembly of claim 1, wherein the anion conducting polymer comprises quaternary ammonium functional groups.

5. The membrane electrode assembly of claim 1, wherein the anion conducting polymer comprises phosphonium groups functional groups.

6. The membrane electrode assembly of claim 1, wherein the anion conducting polymer comprises alkyl or a piperidine side chain configured between a functional group and a backbone of the anion conducting polymer.

7. The membrane electrode assembly of claim 1, wherein the anion conducting polymer comprises a backbone selected from the group consisting of: poly(arylene), poly(phenylene) and poly(styrene).

8. The membrane electrode assembly of claim 1, wherein the anion conducting polymer comprises:

- quaternary ammonium functional groups,
- a backbone selected from the group consisting of: poly(arylene), poly(phenylene) and poly(styrene); and
- an alkyl or piperidine side chain configured between a functional group and a backbone of the anion conducting polymer.

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9. The membrane electrode assembly of claim 1, wherein the anion conducting layer is a composite anion conducting layer further comprising a support material attached to the anion conducting polymer.

10. The membrane electrode assembly of claim 9 wherein the support material has a porosity of 55% to 90%.

11. The membrane electrode assembly of claim 1 wherein the anion conducting layer has a thickness between 5 and 50 microns.

12. The membrane electrode assembly of claim 1, wherein the working fluid comprises ammonia.

13. A refrigeration system defining a closed loop that contains a working fluid, wherein at least part of the working fluid is circulated through the closed loop, the refrigeration system comprising:

- a) an electrochemical compressor system comprising an electrochemical compressor comprising the membrane electrode assembly of claim 1;
- b) a condenser; and
- c) an evaporator.

14. The refrigeration system of claim 13, wherein the working fluid comprises carbon dioxide; wherein the carbon dioxide is in equilibrium with bicarbonate ions and wherein the bicarbonate ions are trans-

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ported through the anion conducting layer from the anode to the cathode and reformed into carbon dioxide at the cathode.

15. The refrigeration system of claim 14, wherein the anion conducting polymer comprises:
quaternary ammonium functional groups,
a backbone selected from the group consisting of: poly(arylene), poly(phenylene) and poly(styrene); and
an alkyl or piperidine side chain configured between a functional group and a backbone of the anion conducting polymer.

16. The refrigeration system of claim 15, wherein the anisotropic anion conducting layer comprise a first anion conducting layer having a first concentration of functional groups and a second anion conducting layer having a second concentration of functional groups and wherein the first anion conducting layer is in contact with the anode and wherein the second anion conducting layer is in contact with the cathode; and

wherein the first concentration of functional groups is at least 1.5 greater than the second concentration of functional groups.

17. The refrigeration system of claim 13, wherein the working fluid comprises ammonia.

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